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(54) **ROTARY DIAL STRUCTURE**

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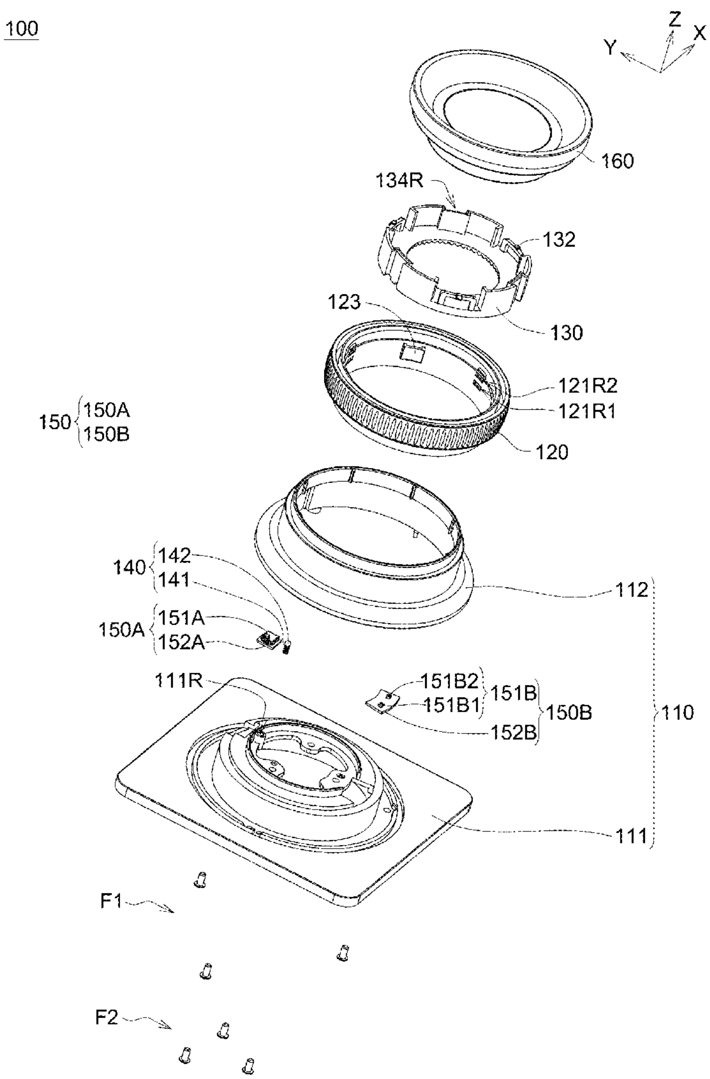
(57) **ABSTRACT**

A rotary dial structure comprises a base, an inner ring component and an outer ring component. The inner ring component is on the base. The outer ring component sleeves onto the inner ring component, the outer ring component and the inner ring component are relatively movable between a first position and a second position, and the outer ring component and the inner ring component are synchronously rotatable at the first position or the second position. The rotary dial structure provided by the present invention is capable of switching modes by moving the outer ring component and adjusting the settings of the switched modes by rotating the outer ring component, thereby achieving a real-time and rapid adjustment.



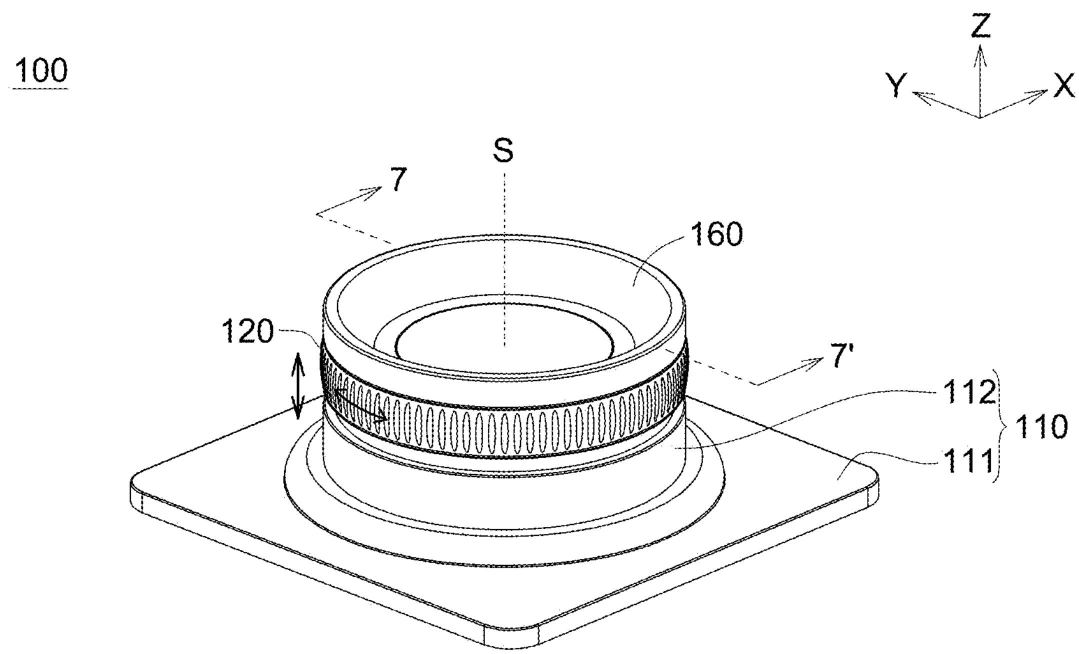


FIG. 1

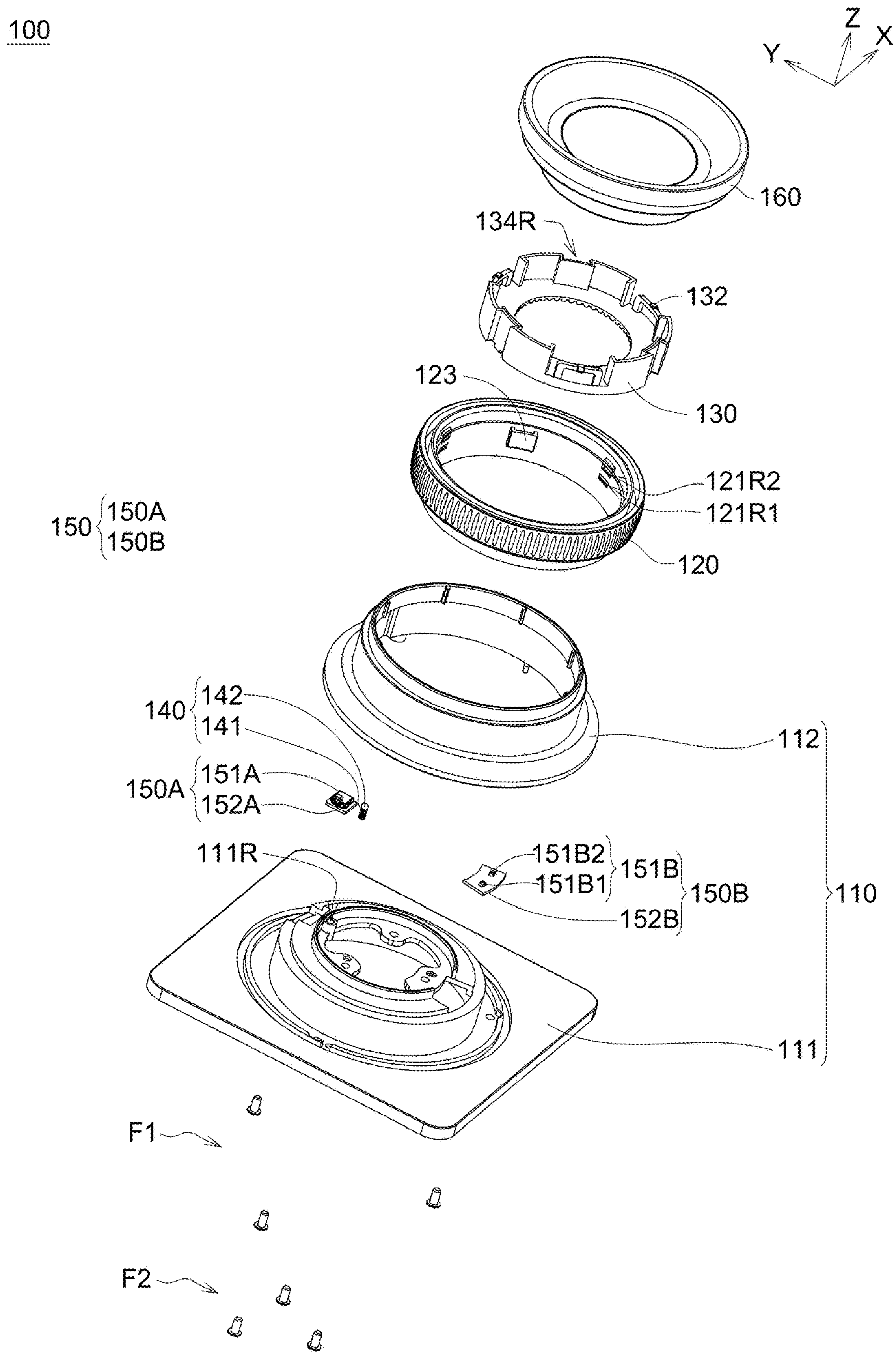

100
Z
Y
X
160
134R
132
123
130
121R2
121R1
120
150
150A
150B
140
142
141
112
150A
151A
152A
111R
151B2
151B1
151B
150B
152B
110
111
F1
F2

FIG. 2

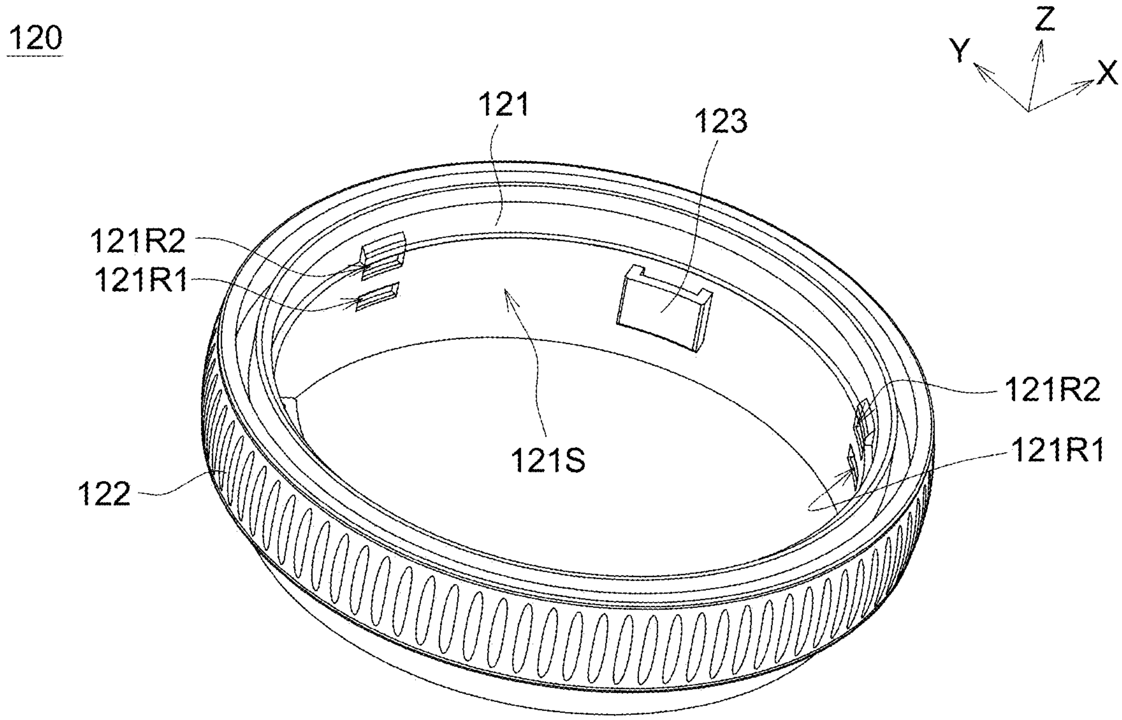

120
Z
Y
X
121
123
121R2
121R1
121R2
121S
121R1
122

FIG. 3

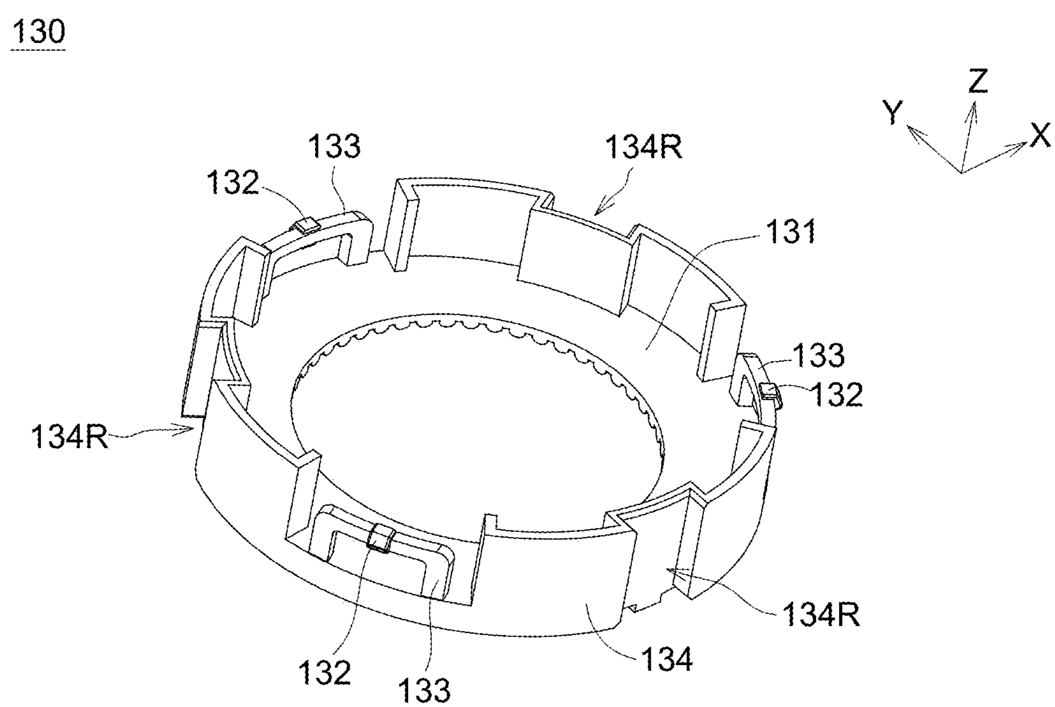

130
Z
Y
X
133
134R
132
131
133
132
134R
134R
134
132
133

FIG. 4

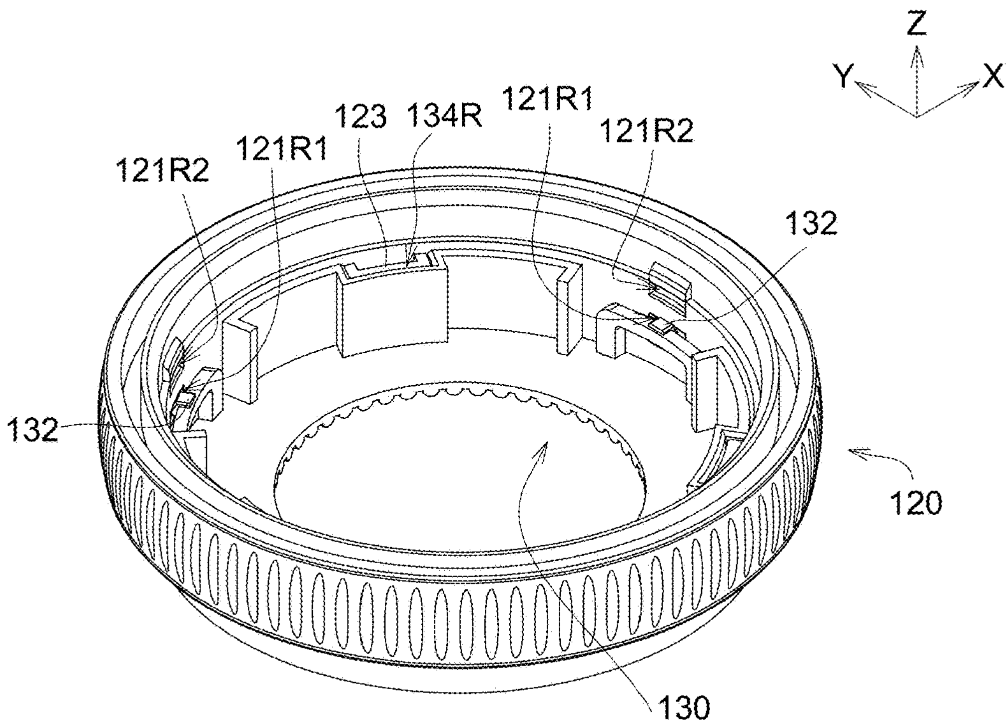

Z
Y
X
123
134R
121R1
121R2
121R2
121R1
132
132
120
130

FIG. 5A

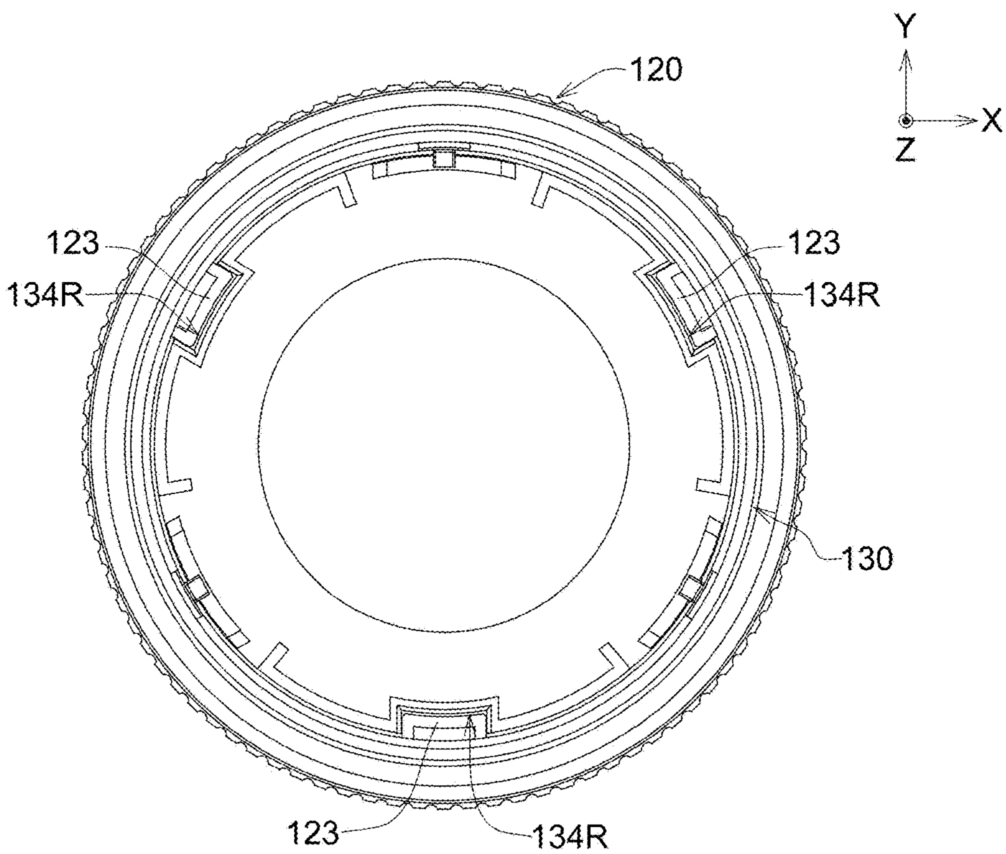

Y
X
Z
120
123
123
134R
134R
130
123
134R

FIG. 5B

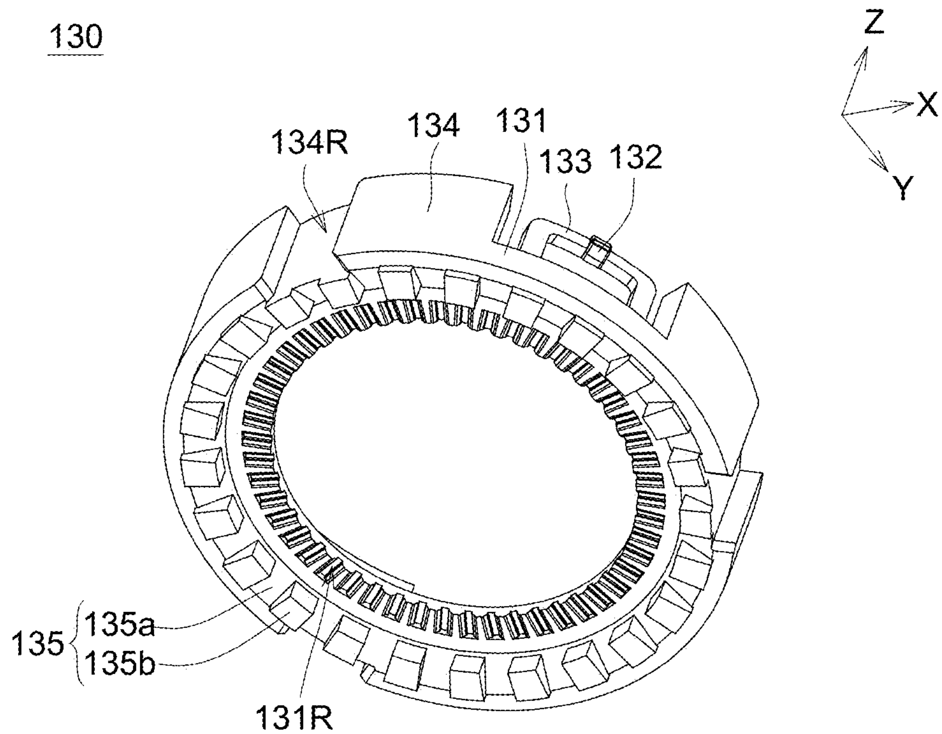


FIG. 6

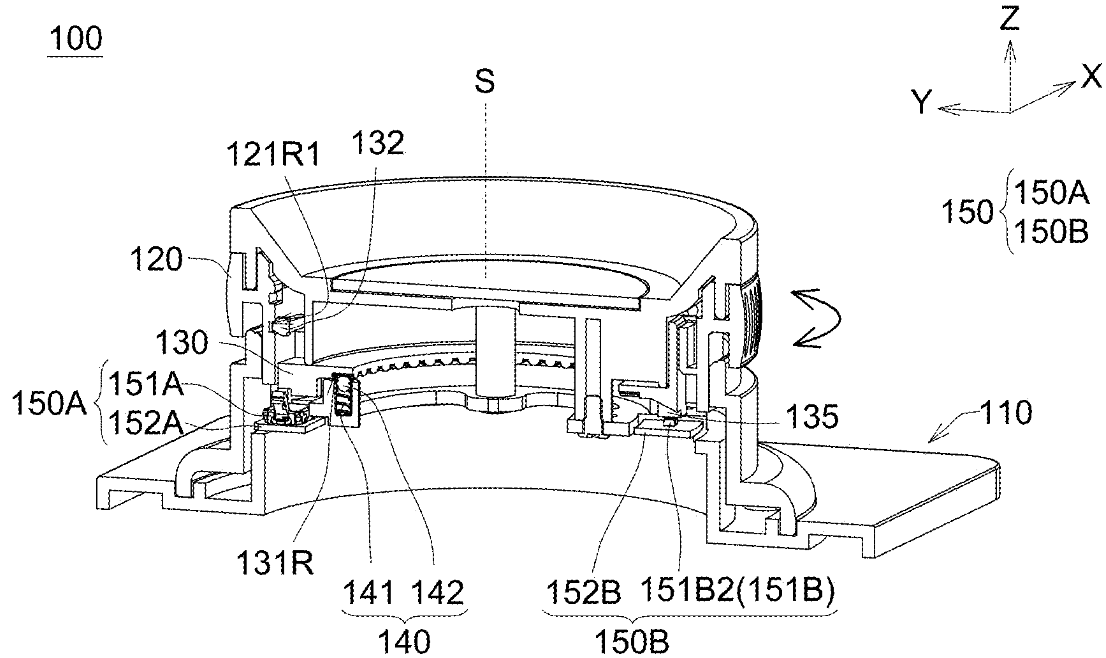

100
S
Z
X
Y
121R1
132
150
150A
150B
120
130
150A
151A
152A
135
110
131R
141
142
140
152B
151B2(151B)
150B

FIG. 7

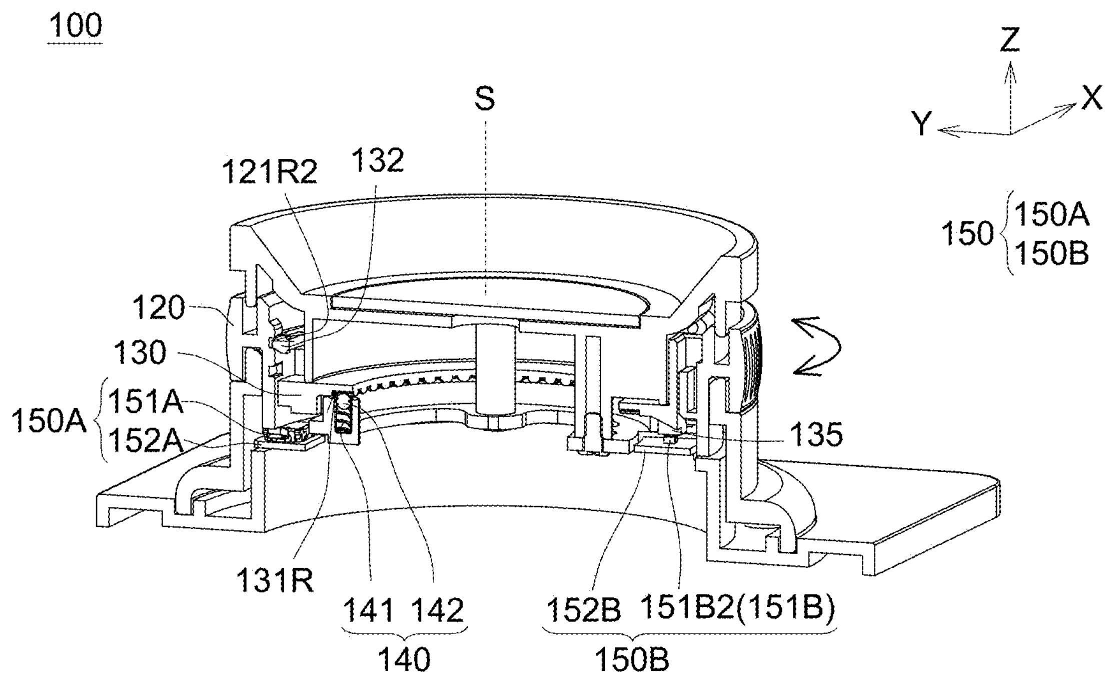

100
S
Z
X
Y
121R2
132
150
150A
150B
120
130
150A
151A
152A
135
131R
141
142
140
152B
151B2(151B)
150B

FIG. 8

ROTARY DIAL STRUCTURE

[0001] This application claims the benefit of the Taiwan Application Serial No. 113202282 filed Mar. 5, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present invention relates in general to a rotary dial structure, and more particularly to a rotary dial structure operable by sliding and rotating.

BACKGROUND

[0003] Currently, electronic devices on the market can utilize different rotary dial structures or screen menus to adjust the operating modes or parameter settings of the electronic devices. When users need to adjust the modes or parameters of the electronic devices, they are required to separately operate the rotary dial structures corresponding to these modes or navigate through the screen menus to make the necessary adjustments. However, such rotary dial structures are unable to provide users with a real-time and efficient way to quickly adjust the settings for different modes of the electronic devices.

SUMMARY

[0004] In view of the shortcomings of the existing technology, the present invention relates to a rotary dial structure that allows mode switching by moving an outer ring component and enables adjustment of the settings for the switched mode by rotating the outer ring component, thereby achieving real-time and rapid adjustments.

[0005] In order to achieve the above purpose, the present invention provides a rotary dial structure comprising a base, an inner ring component, and an outer ring component. The inner ring component is on the base. The outer ring component is sleeved onto the inner ring component. The outer ring component and the inner ring component are relatively movable between a first position and a second position. The outer ring component and the inner ring component are synchronously rotatable at the first position or the second position.

[0006] The above and other aspects of the present invention will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment(s). The following description is made with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a three-dimensional (3D) view of a rotary dial structure according to an embodiment of the present invention.

[0008] FIG. **2** is an exploded view of the rotary dial structure of FIG. **1**.

[0009] FIG. **3** is a 3D view of the outer ring component of FIG. **2**.

[0010] FIG. **4** is a 3D view of the inner ring component of FIG. **2**.

[0011] FIG. **5**A is an assembly drawing of the outer ring component and the inner ring component of FIG. **2**.

[0012] FIG. **5**B is an assembly drawing of the outer ring component and the inner ring component of FIG. **2** from a top view.

[0013] FIG. **6** is a 3D view of the inner ring component of FIG. **3** from another viewpoint.

[0014] FIG. **7** is a 3D cross-sectional schematic view of the rotary dial structure of FIG. **1** along the section line **7-7'**, illustrating the outer ring component at a first position.

[0015] FIG. **8** is a 3D cross-sectional schematic view of the rotary dial structure, illustrating the outer ring component at a second position.

DETAILED DESCRIPTION

[0016] Each embodiment of the present invention will be described in detail below, and the drawings will be used as illustrations. In addition to these detailed descriptions, the present invention can also be widely implemented in other embodiments. Easy substitutions, modifications, and equivalent changes of any of the embodiments are included in the scope of the present invention, and the scope of the present invention depends on the subsequent claims. In the description of the specification, many specific details and implementation examples are provided to enable readers to have a more complete understanding of the present invention; however, these specific details and implementation examples should not be regarded as limitations of the present invention. In addition, well-known steps or components are not described in detail to avoid unnecessary limitations of the present invention.

[0017] FIG. **1** is a three-dimensional (3D) view of a rotary dial structure **100** according to an embodiment of the present invention. FIG. **2** is an exploded view of the rotary dial structure **100** of FIG. **1**.

[0018] Referring to FIGS. **1** and **2**, the rotary dial structure **100** at least comprises a base **110**, an outer ring component **120** and an inner ring component **130**. The inner ring component **130** is disposed on the base **110**. The outer ring component **120** is sleeved onto the inner ring component **130**, so that the outer ring component **120** surrounds the inner ring component **130**.

[0019] In a specific embodiment, the rotary dial structure **100** may be applied to electronic devices with light lenses, such as cameras, camcorders, projectors or sights. The base **110** may include a supporting seat **111** and a lens barrel **112**, and the supporting seat **111** and the lens barrel **112** may be fixed relative to each other by a fastening element F**1**. The rotary dial structure **100** may further comprise a lens cover **160**, and a fastening element F**2** may pass through the supporting seat **111** to fix the lens cover **160** to the supporting seat **111**. Additionally, a snap-fit structure or adhesive material, which is not limited by the present invention, may be used to fix the supporting seat **111** to the lens barrel **112**, as well as the lens cover **160** to the supporting seat **111**.

[0020] As shown in FIGS. **1** and **2**, the outer ring component **120** has a rotation axis S, and the rotation axis S is parallel to the Z-axis. The outer ring component **120** may be disposed in the space between the lens barrel **112** and the lens cover **160**, and the outer ring component **120** may move linearly along the direction of the rotation axis S between a first position and a second position. When the outer ring component **120** is at the first position or the second position, the outer ring component **120** rotates synchronously with the inner ring component **130** around the rotation axis S.

[0021] FIG. **3** is a 3D view of the outer ring component **120** of FIG. **2**. FIG. **4** is a 3D view of the inner ring component **130** of FIG. **2**. FIG. **5**A is an assembly drawing of the outer ring component **120** and the inner ring component **130** of FIG. **2**. FIG. **5**B is an assembly drawing of the outer ring component **120** and the inner ring component **130** of FIG. **2** from a top view.

[0022] Please refer to FIG. **3**, the outer ring component **120** may include an annular body **121**, an operating part **122** and at least two first positioning parts **121**R1 and **121**R2. The operating part **122** is formed on the outer circumferential surface of the annular body **121** and may be used by a user to slide or rotate the outer ring component **120**. The first positioning parts **121** R1 and **121**R2 are on the inner circumferential surface **121**S of the annular body **121** and are along the Z-axis direction. The first positioning parts **121** R1 and **121** R2 may be recessed structures.

[0023] Please refer to FIG. **4**, the inner ring component **130** may include a base part **131**, a second positioning part **132**, an elastic part **133** and a sidewall part **134**. The elastic part **133** and the sidewall part **134** are formed on the base part **131** and are located at the periphery of the base part **131**. The second positioning part **132** is on one side of the elastic part **133** along the Z-axis direction relative to the base part **131**. The second positioning part **132** may be a protruding structure. While the elastic part **133** is subjected to a radial pushing force or a radial pressing force, it drives the second positioning part **132** to move in response to the applied force.

[0024] In a specific embodiment, the shape of the elastic part **133** may be an inverted U-shape, but the present invention is not limited thereto. For example, the elastic part **133** may also be a straight-line shape. Alternatively, in other embodiments, the elastic part **133** may be omitted, and the second positioning part **132** may be formed on the sidewall part **134** and inherently possess a certain degree of elasticity to move in response to a radial pushing force or a pressing force.

[0025] Please refer to FIGS. **3**, **4**, and **5**A, while the outer ring component **120** and the inner ring component **130** are assembled relative to each other, the second positioning part **132** engages with one of the first positioning parts **121** R1 and **121** R2. Taking FIG. **5**A as an example, the second positioning part **132** is engaged with the first positioning part **121** R1, at which point the outer ring component **120** is at the first position. While the outer ring component **120** slides in the negative Z-axis direction, the second positioning part **132** moves and engages with the first positioning part **121** R2 due to the elasticity of the elastic part **133**, at which point the outer ring component **120** is at the second position. Therefore, the outer ring component **120** is at the first position or the second position by engaging the second positioning part **132** with one of the at least two first positioning parts **121** R1 or **121** R2.

[0026] In another embodiment, the second positioning part **132** and the at least two first positioning parts **121**R1 and **121**R2 may be in a reversed configuration. For instance, the second positioning part **132** may be on the outer ring component **120**, while the first positioning parts **121** R1 and **121**R2 may be on the inner ring component **130**. Specifically, the outer ring component **120** may include a protruding structure, and the inner ring component **130** may include a plurality of recessed structures. Alternatively, the outer ring component **120** may include a plurality of recessed structures, while the inner ring component **130** may include a protruding structure. Through the relative movement of the outer ring component **120** and the inner ring component **130** in the rotation axis S, the protruding structure engages with one of the plurality of recessed structures.

[0027] Please refer to FIG. **3**, FIG. **4**, and FIG. **5**B, the outer ring component **120** further includes a protrusion **123**, and the protrusion **123** protrudes from the inner circumferential surface **121**S of the annular body **121**. The inner ring component **130** includes a recess **134**R, and the recess **134**R is recessed into the outer surface of the sidewall part **134** and penetrates through the inner ring component **130** along the Z-axis direction, from the positive Z-axis side to the negative Z-axis side. Accordingly, the recess **134**R has open ends on both the positive and negative Z-axis sides. While the outer ring component **120** and the inner ring component **130** are assembled, the protrusion **123** is within the recess **134**R, and the shape of the protrusion **123** corresponds to that of the recess **134**R. Since the recess **134**R is formed along the Z-axis direction, the outer ring component **120** may move relative to the inner ring component **130** along the Z-axis direction through the protrusion **123** and the recess **134**R. Moreover, while the outer ring component **120** rotates, the outer ring component **120** drives the inner ring component **130** to rotate synchronously through the protrusion **123** and the recess **134**R.

[0028] Please refer to FIG. **1** and FIG. **2**, the rotary dial structure **100** further comprises an elastic movable component **140**, and the elastic movable component **140** is substantially parallel to the rotation axis S. While the inner ring component **130** is driven by the outer ring component **120** and rotates synchronously with the outer ring component **120** around the rotation axis S, the elastic movable component **140** provides tactile feedback with a detent feeling.

[0029] The elastic movable component **140** is between the base **110** and the inner ring component **130**. In a specific embodiment, the supporting seat **111** of the base **110** may have a groove **111**R. The elastic movable component **140** is on the supporting seat **111** through the groove **111**R. The elastic movable component **140** may include an elastic element **141** and an embedding element **142**. One end of the elastic element **141** is connected to the supporting seat **111**, and another end of the elastic element **141** is connected to the embedding element **142**. The elastic element **141** may be implemented as a compression spring.

[0030] FIG. **6** is a 3D view of the inner ring component **130** of FIG. **3** from another viewpoint. Please refer to FIG. **2** and FIG. **6**, the inner ring component **130** further includes a plurality of embedding grooves **131**R. These embedding grooves **131**R are along the rotational direction of the inner ring component **130** and are recessed into the base part **131** from its bottom surface in the positive Z-axis direction. The groove **111**R of the supporting seat **111** corresponds to one of the embedding grooves **131**R. While the elastic movable component **140** is inserted into the groove **111**R of the supporting seat **111**, the embedding element **142** protrudes from the supporting seat **111** and is against one of the embedding grooves **131**R and the elastic element **141**.

[0031] During the synchronous rotation of the outer ring component **120** and the inner ring component **130**, the base part **131** of the inner ring component **130** generates a force that drives the embedding element **142** to move along the Z-axis, thereby compressing the elastic element **141**. In the process where the outer ring component **120** and the inner ring component **130** continue to rotate, causing the embedding element **142** to gradually move away from the embedding groove **131**R that it is originally against and to approach the next adjacent embedding groove **131**R, the embedding element **142** is against the next adjacent embedding groove **131**R by the restoring force of the elastic element **141**. Thus, the embedding element **142** sequentially passes through these embedding grooves **131**R, thereby providing tactile feedback in the form of a detent feeling.

[0032] Please refer to FIG. **1** and FIG. **2**, the rotary dial structure **100** may further comprise a signal sensing module **150**. While the outer ring component **120** is at the first position and rotates synchronously with the inner ring component **130**, the signal sensing module **150** detects an operation signal of a first mode. While the outer ring component **120** is at the second position and rotates synchronously with the inner ring component **130**, the signal sensing module **150** detects an operation signal of a second mode.

[0033] The signal sensing module **150** may include a switching module **150**A and a light sensing module **150**B. The switching module **150**A is configured to detect whether the current mode of the rotary dial structure **100** is the first mode or the second mode. The switching module **150**A may have a switch **151**A and a circuit board **152**A, and the switch **151**A is on the circuit board **152**A. While the outer ring component **120** is at the first position, the switch **151**A is not triggered by the outer ring component **120** and outputs an activation signal for the first mode. Conversely, while the outer ring component **120** is at the second position, the outer ring component **120** triggers the switch **151**A, and the switch **151**A outputs an activation signal for the second mode.

[0034] The light sensing module **150**B is configured to detect rotation signals of the inner ring component **130**. The light sensing module **150**B may include a light sensor **151**B and a circuit board **152**B, and the light sensor **151**B is on the circuit board **152**B. Please refer to FIG. **2** and FIG. **6**, the inner ring component **130** may include a light-detectable stripe structure **135**. The light-detectable stripe structure **135** is formed on the bottom surface of the base part **131** and is along the rotational direction of the inner ring component **130**. While the inner ring component **130** is driven by the outer ring component **120** to rotate synchronously with the outer ring component **120**, the light sensor **151**B detects changes in the light-detectable stripe structure **135** and output a rotation signal. In a specific embodiment, the light-detectable stripe structure **135** may include alternating bright and dark sensing stripes **135***a* and **135***b*. The light sensor **151**B may include a first light sensing unit **151**B1 and a second light sensing unit **151**B2. The first light sensing unit **151**B1 and the second light sensing unit **151**B2 respectively detect changes in the sensing stripes **135***a* and **135***b* and output corresponding rotation signals.

[0035] FIG. **7** is a 3D cross-sectional view of the rotary dial structure **100** of FIG. **1** along the section line **7-7**', illustrating the outer ring component **120** at the first position. FIG. **8** is a 3D cross-sectional view of the rotary dial structure **100**, illustrating the outer ring component **120** at the second position.

[0036] Please refer to FIG. **7**, the second positioning part **132** is in the first positioning part **121**R**1** to position the outer ring component **120** at the first position. At this point, the switch **151**A is not triggered by the outer ring component **120**, so the switch **151**A outputs an activation signal for the first mode. While the outer ring component **120** and the inner ring component **130** rotate synchronously around the rotation axis S, the elastic movable component **140** provides tactile feedback with a detent feeling to the user. Furthermore, the light sensing module **150**B detects changes in the light-detectable stripe structure **135**, such as detecting the rotational direction of the inner ring component **130** as clockwise or counterclockwise, and/or detecting the rotational speed of the inner ring component **130**, thus outputting corresponding rotation signals. As a result, while the outer ring component **120** is at the first position and rotates synchronously with the inner ring component **130**, the signal sensing module **150** detects the operation signal of the first mode.

[0037] Please refer to FIG. **8**, where the second positioning part **132** is in the first positioning part **121** R**2** to position the outer ring component **120** at the second position. At this point, the outer ring component **120** is pressed down and triggers the switch **151**A, and then the switch **151**A outputs an activation signal for the second mode. As the outer ring component **120** and the inner ring component **130** rotate synchronously around the rotation axis S, the elastic movable component **140** provides tactile feedback with a detent feeling to the user. Furthermore, the light sensing module **150**B detects changes in the light-detectable stripe structure **135**, such as detecting the rotational direction of the inner ring component **130** as clockwise or counterclockwise, and/or detecting the rotational speed of the inner ring component **130**, thus outputting the corresponding rotation signal. As a result, while the outer ring component **120** is at the second position and rotates synchronously with the inner ring component **130**, the signal sensing module **150** detects the operation signal of the second mode.

[0038] In an embodiment, the rotary dial structure **100** may further comprise a control module (not shown). The control module is electrically connected to the signal sensing module **150** and switches between the first and second modes based on the activation signals output by the switching module **150**A. Additionally, the control module adjusts the functions corresponding to the first and second modes based on the rotation signals output by the light-sensing module **150**B. These adjustments may include changes to camera settings, such as aperture, focal length, depth of field, camera image style, and so on.

[0039] It will be apparent to those skilled in the art that various modifications and variations may be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplars only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

What is claimed is:

1. A rotary dial structure, comprising:

a base;

an inner ring component, being on the base; and

an outer ring component, sleeving onto the inner ring component, the outer ring component and the inner ring component are relatively movable between a first position and a second position, and the outer ring component and the inner ring component are synchronously rotatable at the first position or the second position.

2. The rotary dial structure according to claim **1**, wherein the rotary dial structure further comprises a signal sensing module, while the outer ring component rotates synchronously with the inner ring component at the first position, the signal sensing module senses an operation signal of a first mode, and/or while the outer ring component rotates synchronously with the inner ring component at the second position, the signal sensing module senses an operation signal of a second mode.

3. The rotary dial structure according to claim **2**, wherein the signal sensing module includes a switching module, and the switching module has a switch.

4. The rotary dial structure according to claim **2**, wherein the signal sensing module includes a light sensor, and the inner ring component includes a light-detectable stripe structure.

5. The rotary dial structure according to claim **1**, wherein the inner ring component includes two first positioning parts and the outer ring component includes a second positioning part, or the inner ring component includes a second positioning part and the outer ring component includes two first positioning parts, the outer ring component is at the first position or the second position by engaging the second positioning part with one of the first positioning parts.

6. The rotary dial structure according to claim **1**, wherein the inner ring component further includes an elastic part and a protruding structure, and the protruding structure is on the elastic part.

7. The rotary dial structure according to claim **1**, wherein the rotary dial structure further comprises an elastic movable component, and the elastic movable component is between the base and the inner ring component.

8. The rotary dial structure according to claim **7**, wherein the elastic movable component includes an elastic element and an embedding element, the inner ring component has a plurality of embedding grooves along a rotational direction of the inner ring component, and the embedding element is against one of the embedding grooves and the elastic element.

9. The rotary dial structure according to claim **1**, wherein the outer ring component and the inner ring component respectively include a protrusion and a recess, the outer ring component is movable relative to or synchronously rotatable with the inner ring component through the protrusion and the recess.

10. A rotary dial structure, comprising:

an inner ring component, including a light-detectable stripe structure,

an outer ring component, sleeving onto the inner ring component, the outer ring component and the inner ring component are relatively movable between a first position and a second position, and the outer ring component and the inner ring component are synchronously rotatable at the first position or the second position; and

a signal sensing module, including a switching module and a light sensor, the switching module outputs an activation signal of a first mode or a second mode based on that the outer ring component is at the first position or the second position; and while the outer ring component and the inner ring component rotate synchronously, the light sensor detects changes in the light-detectable stripe structure and outputs a rotation signal.

11. The rotary dial structure according to claim **10**, wherein the switching module has a switch, while the outer ring component is at the first position, the switch outputs an activation signal for the first mode; and while the outer ring component is at the second position, the switch outputs an activation signal for the second mode.

12. The rotary dial structure according to claim **10**, wherein the light-detectable stripe structure is along a rotational direction of the inner ring component.

13. The rotary dial structure according to claim **10**, wherein the inner ring component includes two first positioning parts and the outer ring component includes a second positioning part, or the inner ring component includes a second positioning part and the outer ring component includes two first positioning parts, the outer ring component is at the first position or the second position by engaging the second positioning part with one of the first positioning parts.

14. The rotary dial structure according to claim **10**, wherein the inner ring component further includes an elastic part and a protruding structure, and the protruding structure is on the elastic part.

15. A rotary dial structure, comprising:

a base;

an inner ring component, being on the base;

an outer ring component, sleeving onto the inner ring component, the outer ring component and the inner ring component are relatively movable between a first position and a second position, and the outer ring component and the inner ring component are synchronously rotatable at the first position or the second position; and

a signal sensing module, detecting an operation signal of a first mode or an operation signal of a second mode based on that the outer ring component and the inner ring component rotate synchronously at the first position or the second position.

16. The rotary dial structure according to claim **15**, wherein the signal sensing module includes a switching module, the switching module has a switch, while the outer ring component is at the first position, the switch outputs an activation signal for the first mode; and while the outer ring component is at the second position, the outer ring component triggers the switch and the switch outputs an activation signal for the second mode.

17. The rotary dial structure according to claim **15**, wherein the signal sensing module includes a light sensor, the inner ring component includes a light-detectable stripe structure, while the outer ring component and the inner ring component rotate synchronously, the light sensor detects changes in the light-detectable stripe structure and outputs a rotation signal.

18. The rotary dial structure according to claim **15**, wherein the inner ring component further includes an elastic part and a protruding structure, and the protruding structure is on the elastic part.

19. The rotary dial structure according to claim **15**, wherein the rotary dial structure further comprises an elastic movable component, and the elastic movable component is between the base and the inner ring component.

20. The rotary dial structure according to claim **19**, wherein the elastic movable component includes an elastic element and an embedding element, the inner ring component has a plurality of embedding grooves along a rotational direction of the inner ring component, and the embedding element is against one of the embedding grooves and the elastic element.

* * * * *